

Hand Calculation: Series and Parallel Resistors

Given values

Network	Source	Resistors
Series	$V_1 = 9\text{ V}$	$R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$
Parallel	$V_2 = 9\text{ V}$	$R_3 = 1\text{ k}\Omega$, $R_4 = 2\text{ k}\Omega$

Series network

The equivalent resistance is

$$[R_{eq,s} = R_1 + R_2 = 1\,\mathrm{k\Omega} + 2\,\mathrm{k\Omega} = 3\,\mathrm{k\Omega}.]$$

The series current is therefore

$$[I_s = \frac{V_1}{R_{eq,s}} = \frac{9\,\mathrm{V}}{3\,\mathrm{k\Omega}} = 3\,\mathrm{mA}.]$$

The voltage drops are

$$[V_{R1} = I_s R_1 = (3\,\mathrm{mA})(1\,\mathrm{k\Omega}) = 3\,\mathrm{V},]$$

$$[V_{R2} = I_s R_2 = (3\,\mathrm{mA})(2\,\mathrm{k\Omega}) = 6\,\mathrm{V}.]$$

The drops satisfy KVL:

$$[V_{R1} + V_{R2} = 3\,\mathrm{V} + 6\,\mathrm{V} = 9\,\mathrm{V} = V_1.]$$

The individual powers and total power are

$$[P_{R1} = I_s^2 R_1 = 9\,\mathrm{mW}, \quad P_{R2} = I_s^2 R_2 = 18\,\mathrm{mW},]$$

$$[P_{series,total} = P_{R1} + P_{R2} = 27\,\mathrm{mW}.]$$

Parallel network

The equivalent resistance is

$$[R_{eq,p} = \frac{R_3 R_4}{R_3 + R_4} = \frac{(1\,\mathrm{k\Omega})(2\,\mathrm{k\Omega})}{1\,\mathrm{k\Omega} + 2\,\mathrm{k\Omega}} = 666.667\,\Omega.]$$

Both branches have the source voltage across them:

$$[V_{R3} = V_{R4} = V_2 = 9\,\mathrm{V}.]$$

The branch currents are

$$[I_{R3} = \frac{9\text{ V}}{1\text{ k}\Omega} = 9\text{ mA}, \text{ and } I_{R4} = \frac{9\text{ V}}{2\text{ k}\Omega} = 4.5\text{ mA}.]$$

The total current is

$$[I_{p,total} = I_{R3} + I_{R4} = 13.5\text{ mA}.]$$

The total power is

$$[P_{parallel,total} = V I_{p,total} = (9\text{ V})(13.5\text{ mA}) = 121.5\text{ mW}.]$$

The branch powers are 81 mW for R3 and 40.5 mW for R4 .

Analytical versus LTspice comparison

Measurement	Analytical	LTspice	Approximate relative error
Series current $I(R1)$	3.000000 mA	3.000000002608 mA	$8.69 \times 10^{-8} \%$
Series drop $V(n_{s0}, n_{s1})$	3.000000 V	3.000000 V	0 % at displayed precision
Series drop $V(n_{s1})$	6.000000 V	6.000000 V	0 % at displayed precision
Parallel current $I(R3)$	9.000000 mA	8.999999961257 mA	$4.30 \times 10^{-7} \%$
Parallel current $I(R4)$	4.500000 mA	4.499999980628 mA	$4.30 \times 10^{-7} \%$
Parallel voltage $V(n_p)$	9.000000 V	9.000000 V	0 % at displayed precision

The tiny differences are numerical solver precision and do not indicate a physical-model discrepancy. The simulation verifies series addition, parallel equivalent resistance, KVL, voltage division, and current division for the selected ideal resistor networks.